IMAGE BRINGUP DOCUMENT

Bridge Board — NVIDIA Jetson Orin Platform

Field	Details
Document Title	Image Bringup Guide — Bridge Board
Platform	NVIDIA Jetson Orin Nano / Orin NX
Host OS	Ubuntu 22.04 / 20.04 LTS
Camera Module	Sony-EV9520L (LVDS Interface)
Prepared By	Oppila Microsystems Pvt. Ltd.
Contact	info@oppila.in
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Date	2024
Classification	Confidential — Client Use Only

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1. Document Overview

This document provides step-by-step instructions for setting up the NVIDIA Jetson Orin platform with an Oppila Microsystems Bridge Board. It covers host PC software configuration, firmware flashing, Bridge Board connection, and GStreamer-based video streaming from a Sony-EV9520L LVDS camera.

1.1 System Architecture

The following block diagram illustrates the high-level system architecture, showing the relationship between the host PC, Jetson developer kit, Bridge Board, and camera module.

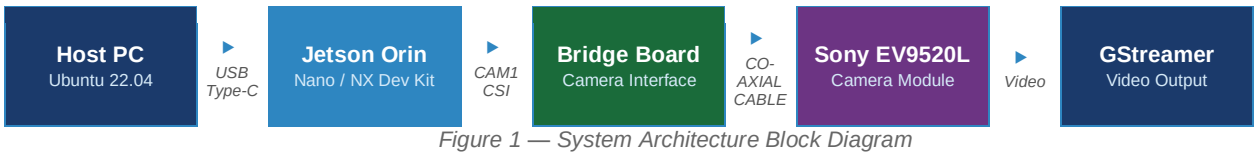

Figure 1 — System Architecture Block Diagram

1.2 Prerequisites

Before proceeding, ensure the following items are available:

- Host PC running Ubuntu 22.04 or 20.04 LTS (64-bit)
- NVIDIA Jetson Orin Nano Developer Kit
- Oppila Microsystems Bridge Board with Sony-EV9520L LVDS camera
- USB Type-C cable for flashing
- Power adapter compatible with the Jetson Developer Kit
- Jumper wire for recovery mode (FC REC ↔ GND)

1.3 Required Files

The following files must be downloaded from the provided Google Drive link before starting:

- rc.local — Runtime configuration script
- ORIN_NX_IMAGE.tar.bz2 — Jetson OS image archive

Download from:
https://drive.google.com/drive/folders/1TDUXYL6gGu0oZq7z41naLT2DZZxhkUus

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2. Host PC Software Setup

This section describes how to prepare the host PC environment on Ubuntu 22.04 or 20.04 LTS. All commands must be executed in a terminal window with administrator (sudo) privileges.

2.1 Prepare Working Directory

1. **Step 1: Download Required Files** — Download rc.local and ORIN_NX_IMAGE.tar.bz2 from the Google Drive link above.
2. **Step 2: Create a Working Directory**

```
mkdir ~/New
```

3. **Step 3: Move the Downloaded Image**

```
mv ~/Downloads/ORIN_NX_IMAGE.tar.bz2 ~/New/
```

4. **Step 4: Navigate to the New Directory**

```
cd ~/New/
```

2.2 Extract the OS Image

5. **Step 5: Extract the Archive**

Run the following command. Extraction may take several minutes.

```
sudo tar -xvJf Linux_for_Tegra_2.tar.bz2
```

Once complete, a directory named Linux_for_Tegra will be created inside ~/New/.

6. **Step 6: Enter into the Extracted Directory**

```
cd Linux_for_Tegra/
```

7. **Step 7: Apply Binaries**

```
sudo ./apply_binaries.sh
```

2.3 Install Dependencies

8. **Step 8: Update Package Lists and Install Required Packages**

```
sudo apt update
sudo apt install -y qemu-user-static sshpass binutils abootimg \   libxml2-utils nfs-kernel-server u-boot-tools \   android-tools-fastboot python3-pip lz4
pip3 install -U pip
```

9. **Step 9: Reboot the Host PC**

```
sudo reboot
```

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3. Hardware Connection & Flashing

This section covers placing the Jetson Developer Kit into recovery mode, connecting it to the host PC, and flashing the OS image.

3.1 Set Recovery Mode

10. **Step 10: Connect Recovery Mode Jumper**

Place a jumper between the FC REC and GND pins on the Jetson Orin Nano Developer Kit, as shown in the image below.

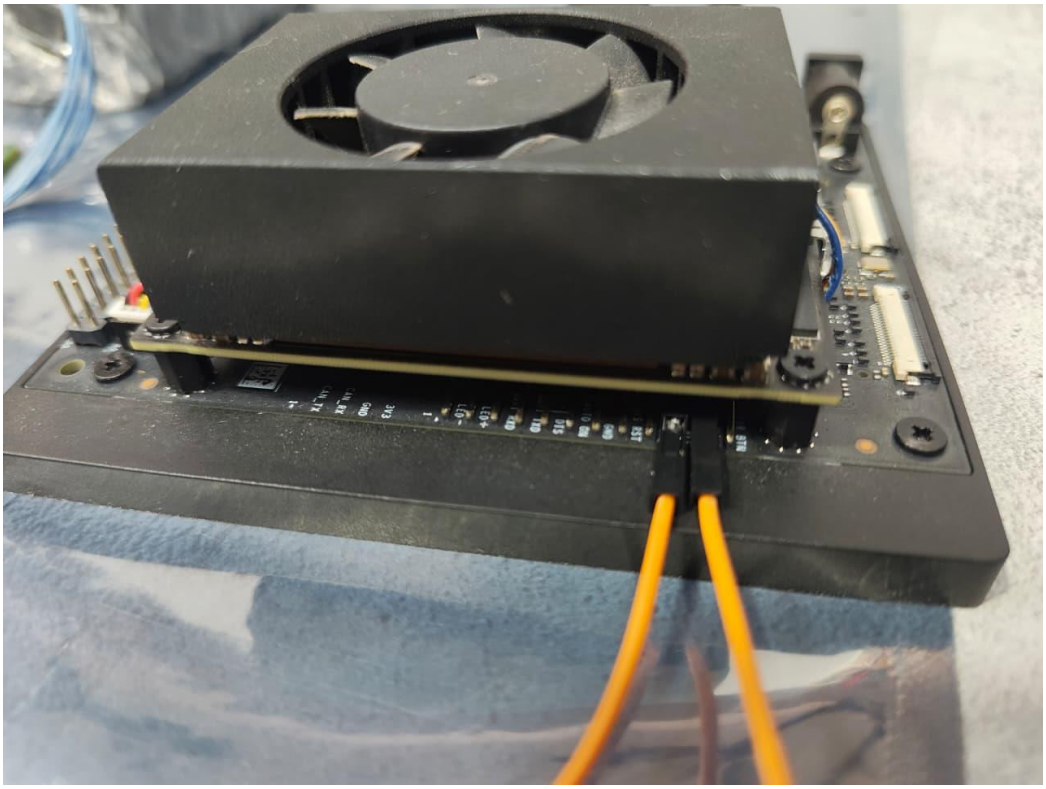

Figure 2 — Jumper placement for FC REC and GND pins (Recovery Mode)

- Power on the developer kit using the provided power adapter.
- Connect a USB Type-C cable from the developer kit to the host PC.

3.2 Verify Device Detection

11. **Step 11: Check Jetson is Detected in Recovery Mode**

```
lsusb
```

Look for an entry such as: NVIDIA Corp. APX

⚠ **Note:** If the device is not visible, try a different USB cable or USB port on the host PC.

3.3 Flash the Jetson

12. **Step 12: Run the Flash Script**

```
sudo ./nvsdkmanager_flash.sh --storage nvme0n1p1
```

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3.4 Complete Setup After Flashing

Once flashing is complete:

- Remove the jumper from the FC REC and GND recovery mode pins.
- Connect the camera module via the Bridge Board.
- Power on the Jetson Orin Nano Developer Kit.

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4. Bridge Board Connection

This section describes how to physically connect the Oppila Microsystems Bridge Board (with Sony-EV9520L LVDS camera) to the Jetson Orin Nano Developer Kit.

4.1 CSI Cable Connection

- Use a CSI cable to connect the camera from the Bridge Board to the CAM1 connector on the Jetson Orin Nano Developer Kit.
- Ensure the cable is seated firmly and the orientation is correct.

Figure 3 — CSI cable connection from Bridge Board to CAM1 connector

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5. Video Configuration & GStreamer Streaming

After the Jetson boots with the Bridge Board connected, follow these steps to configure the camera interface and begin video streaming.

5.1 Transfer Configuration File

13. **Step 1: Copy rc.local to Jetson** — Transfer the rc.local file from your PC to the home directory of the Jetson Orin Nano Developer Kit using a USB drive or SCP.

5.2 Configure LVDS Camera Pins

This is a one-time configuration required after each new flash. It is not required on every boot.

14. **Step 2: Open Terminal on the Jetson** — Press Ctrl + Alt + T on the Jetson to open a terminal window.
15. **Step 3: Run the Jetson-IO Configuration Tool**

```
sudo /opt/nvidia/jetson-io/jetson-io.py
```

Figure 4 — Jetson-IO configuration tool startup

5.3 Pin Configuration Steps

16. **Step 4: Navigate the Configuration Menu**

Follow the menu selections in order:

- Configure Jetson 24-pin CSI Connector
- Configure for compatible hardware
- Camera IMX219-C
- Save pin changes
- Save and reboot to reconfigure pins

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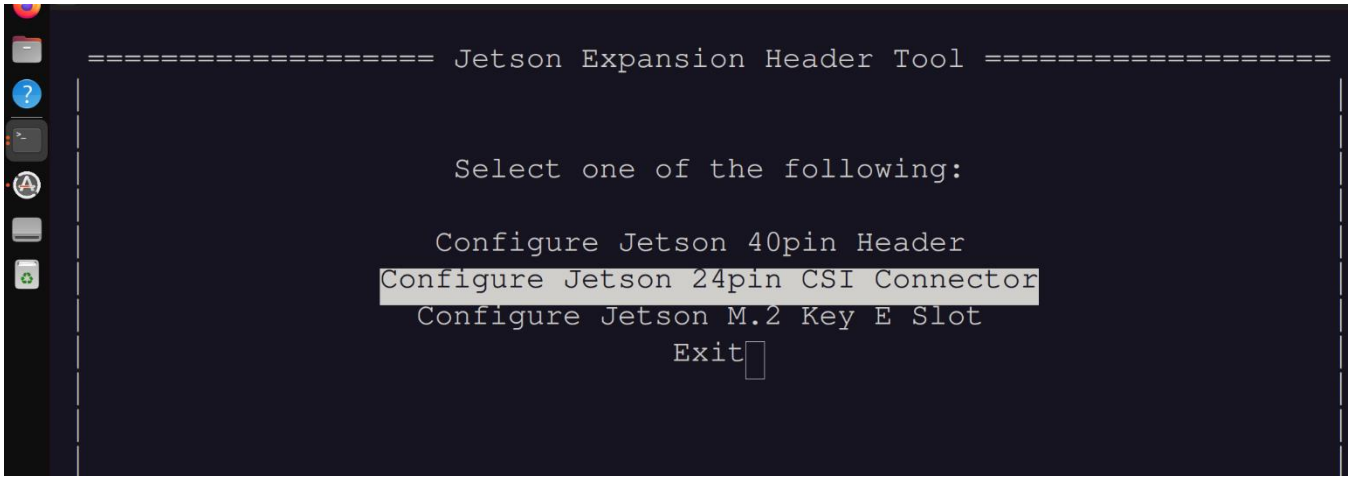

Figure 5 — Select "Configure Jetson 24-pin CSI Connector"

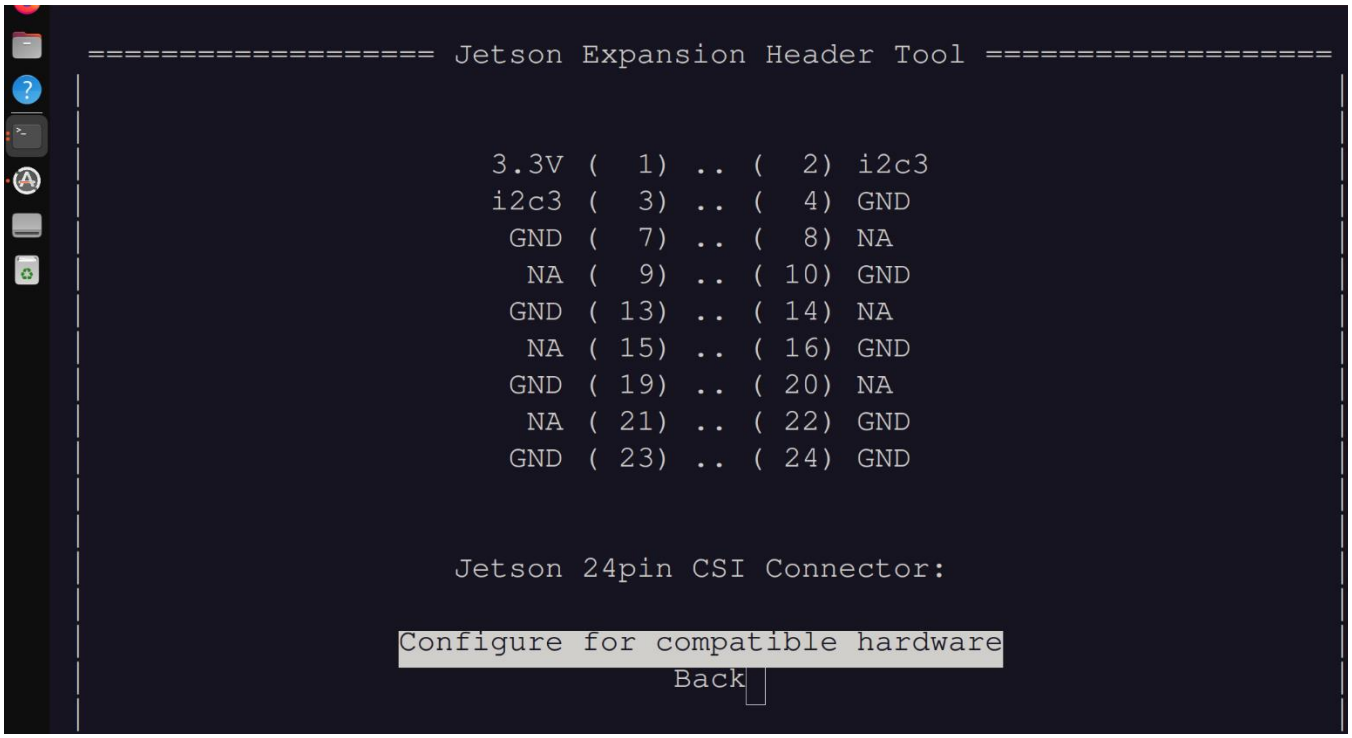

Figure 6 — Select "Configure for compatible hardware"

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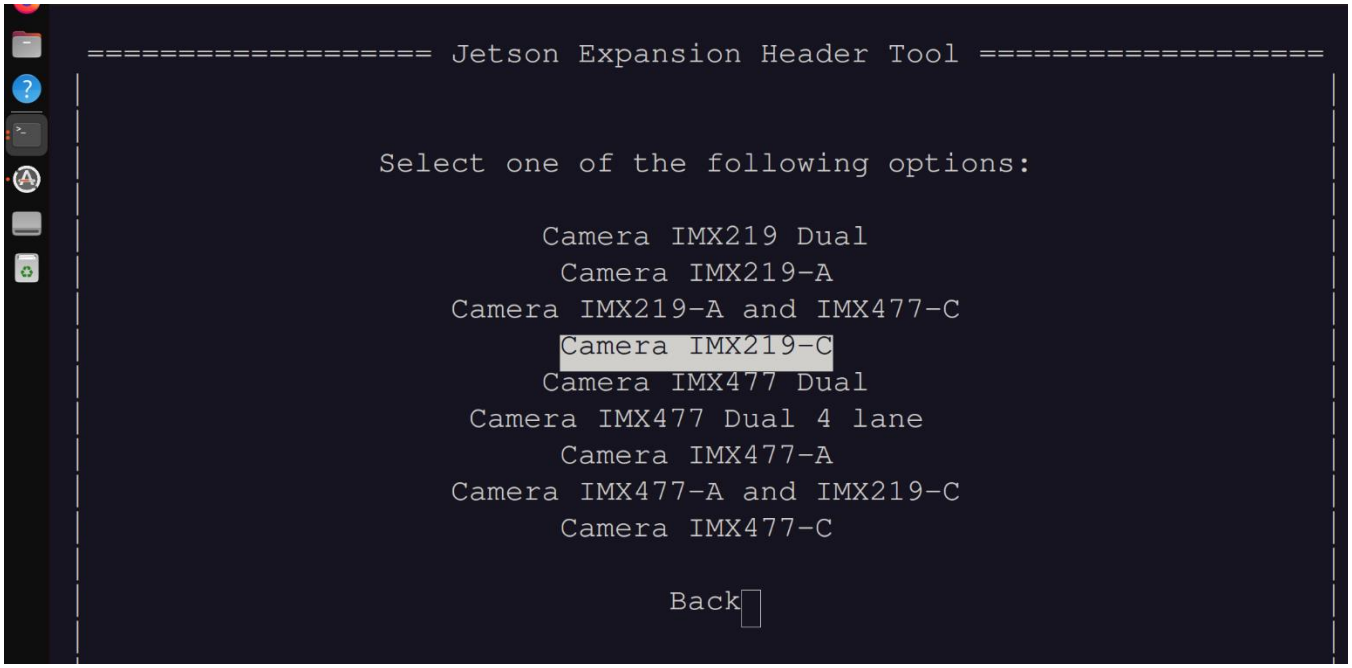

Figure 7 — Select "Camera IMX219-C"

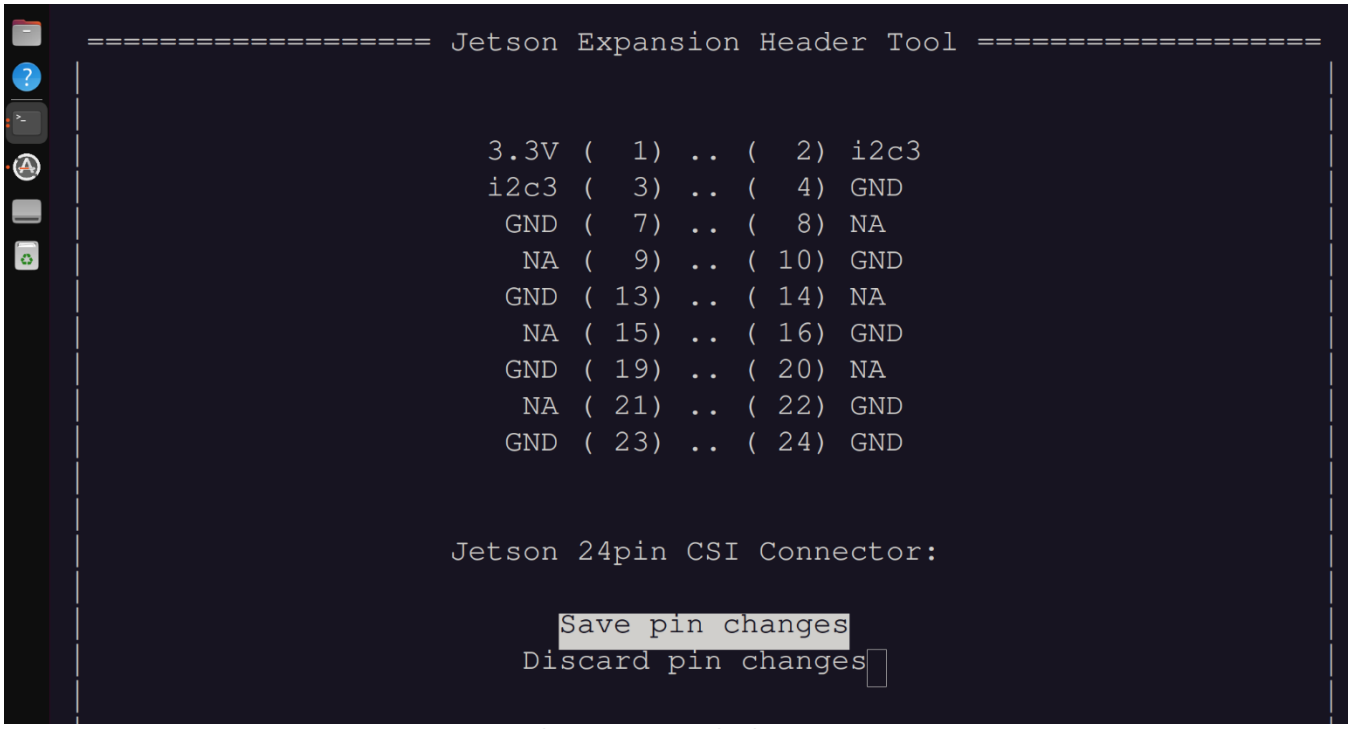

Figure 8 — Save pin changes

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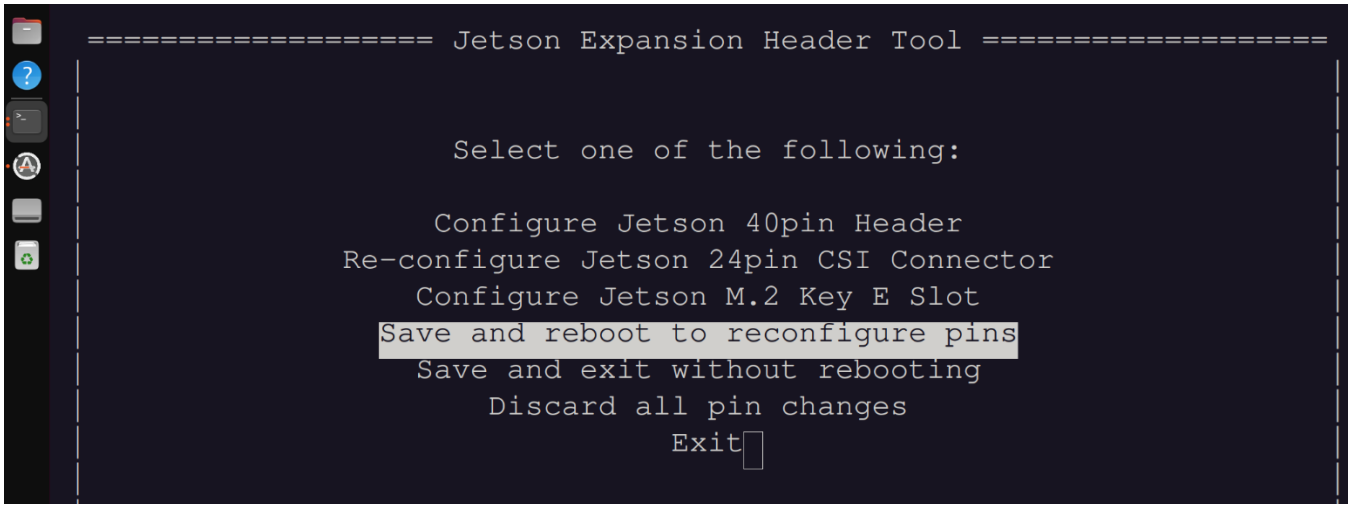

Figure 9 — Save and reboot to reconfigure pins

17. **Step 5: Press Any Key to Reboot** — The system will reboot and reconfigure the MIPI pins for the LVDS camera.

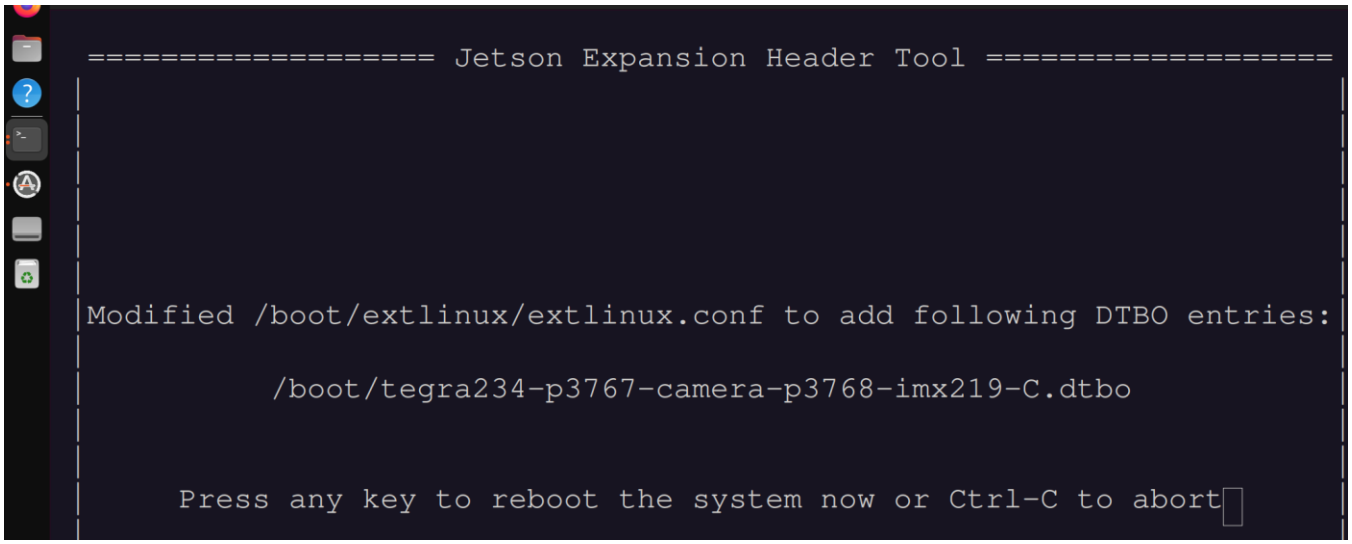

Figure 10 — Reboot confirmation screen

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6. GStreamer Video Streaming

After the system reboots and you log back in, follow these steps to start video streaming.

6.1 Boost Clocks & Enable Debug

18. **Step 6: Open a Terminal and Run rc.local**

```
sudo ./rc.local
```

6.2 Verify Video Node

19. **Step 7: Check the Video Device Node**

```
ls /dev/video*
```

The output should display /dev/video0. If the node is present, the driver has loaded successfully and the camera is ready.

6.3 Start GStreamer Pipeline

20. **Step 8: Run the GStreamer Command**

```
gst-launch-1.0 v4l2src device=/dev/video0 \   ! video/x-raw,format=UYVY \   ! videoconvert \
! xvimagesink
```

If the pipeline starts successfully, a video window will appear displaying the live camera feed.

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7. Troubleshooting

If you encounter issues during any stage, refer to the guidance below.

7.1 Common Issues

Issue	Resolution
Jetson not detected via lsusb	Try a different USB cable or USB port. Ensure the jumper is correctly placed on FC REC and GND.
Video node /dev/video0 not found	Ensure the LVDS pin configuration was completed using jetson-io.py. Re-run the configuration if needed.
Black screen or no output from GStreamer	Verify the CSI cable is connected to CAM1. Check that rc.local was executed successfully.
Flash script fails	Ensure all dependency packages are installed. Re-run apt install and try again.

7.2 Debug Logging

To view kernel-level debug messages related to the LT9211C driver:

```
sudo dmesg | grep -i lt9211c
```

Review the output for any error or warning messages. These logs can help identify initialization failures or communication errors.

7.3 Contact Support

If the issue persists after following all troubleshooting steps, please contact Oppila Microsystems technical support:

Email: info@oppila.in | **Website:** www.oppila.in

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8. Document Revision History

Version	Description
1.0	Initial release — Image bringup guide for Bridge Board

— End of Document —

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